📄 Project Title

Smart Irrigation System Using ESP32 with Web Control and Soil Moisture Sensor

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1. 📌 Introduction

Agriculture is the backbone of many countries, and proper irrigation is essential for good crop yield. Traditional irrigation systems often waste water due to manual control.

This project introduces an **IoT-based Smart Irrigation System** using ESP32 that automatically controls water flow based on soil moisture level and also allows remote control through a web interface.

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2. 🎯 Objectives

- To automate irrigation based on soil moisture level
- To reduce water wastage
- To provide manual and automatic control modes
- To enable remote control via WiFi web server
- To display system status on LCD

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3. □ Components Required

Component Name	Quantity
ESP32	1
Soil Moisture Sensor	1
Solenoid Valve / Relay Module	1
LCD Display (16x2 I2C)	1
Jumper Wires	As required
Power Supply	1

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4. ⚙□ Working Principle

1. Soil Moisture Detection

The soil moisture sensor measures the water content in the soil. The analog value is converted into percentage (0–100%).

- Dry soil → Low value
- Wet soil → High value

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2. Automatic Control Mode

When AUTO mode is enabled:

- Moisture < 30% → Pump ON (soil is dry)
- Moisture > 80% → Pump OFF (soil is wet)

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3. Manual Control Mode

In MANUAL mode:

- User can turn ON/OFF the motor using web buttons

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4. Web Server Control

ESP32 creates a WiFi hotspot:

- User can access dashboard via browser
- Real-time moisture monitoring
- Mode switching (AUTO / MANUAL)
- Motor ON/OFF control

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5. LCD Display

Shows live system status:

- Mode (AUTO / MANUAL)
- Soil moisture percentage
- Motor status (ON / OFF)

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5. 🔌 Pin Configuration

Device	ESP32 Pin
Soil Moisture Sensor	GPIO 34
Solenoid / Relay	GPIO 26
LCD I2C SDA	GPIO 21
LCD I2C SCL	GPIO 22

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6. 🌐 Web Interface Features

- Real-time soil moisture monitoring
- AUTO / MANUAL mode switching
- Motor ON/OFF control buttons
- Auto-refresh data every 1 second
- Mobile-friendly UI

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7. 📟 LCD Output Example

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Mode: AUTO
M: 45%
Mot: ON
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or

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Mode: MANUAL
M: 70%
Mot: OFF
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8. 🔄 System Workflow

1. ESP32 starts and creates WiFi hotspot

2. Soil moisture sensor reads data
3. Data converted into percentage
4. System checks AUTO/MANUAL mode
5. Motor controlled automatically or manually
6. Data displayed on LCD and Web Dashboard

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9. 🚀 Features

✅ Automatic irrigation control
✅ Manual override system
✅ Real-time web monitoring
✅ LCD display output
✅ Low-cost IoT solution
✅ Water saving system

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10. ⚠️ Limitations

- Moisture sensor calibration required
- Not suitable for large industrial farms without scaling
- Requires stable power supply

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11. 💡 Future Improvements

- Add cloud monitoring (Firebase / IoT platform)
- Mobile app control
- Smart weather-based irrigation
- Solar-powered system
- Multiple zone irrigation control

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12. 🏁 Conclusion

This Smart Irrigation System provides an efficient and automated solution for water management in agriculture. By using ESP32 and IoT technology, it reduces manual effort and helps conserve water while improving crop productivity.